TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT I : INTRODUCTION TO TOOL DESIGN

Syllabus (as given)

Introduction - Tool Engineering - Tool Classifications - Tool Design Objectives - Tool Design in manufacturing - Challenges and requirements - Standards in tool design - Tool drawings - Surface finish - Fits and Tolerances - Tooling Materials - Ferrous and Non ferrous Tooling Materials - Carbides, Ceramics and Diamond - Non metallic tool materials - Designing with relation to heat treatment.

1. Introduction to Tool Design

1.1 What is a tool?

- A tool is any device (cutter, die, jig, fixture, gauge) used to cut, form, shape, hold or inspect a workpiece in manufacturing.
- Tool design is the process of designing and developing the tools, methods and techniques needed to improve manufacturing efficiency and productivity.
- It is a specialised branch of production engineering - it joins the product drawing to the actual production on the shop floor.
- The person who does it is the tool designer ; the person who makes the tool is the tool maker .

Why tooling is needed

- To make parts interchangeable and of uniform quality
- To produce large quantities quickly (mass production)
- To reduce the skill needed from the operator
- To reduce cost per piece and inspection time

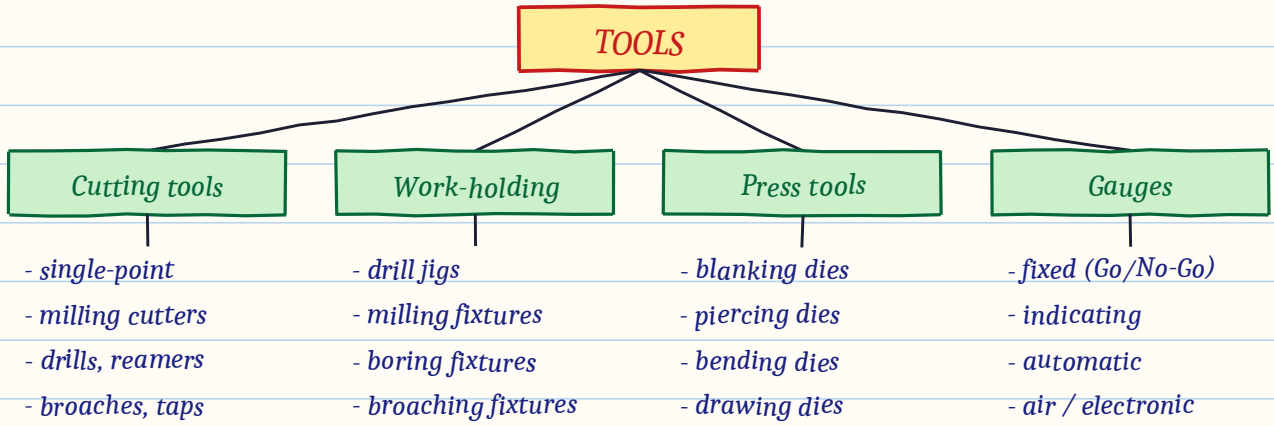
1.2 Tool Engineering

- Tool engineering is the branch of engineering concerned with planning, designing, making and managing all the tooling needed for production.

- It covers: tool design, tool planning, tool manufacture, tool inspection, tool control (store, maintenance, regrinding) .
- Functions of the tool engineer:
  - Study the part drawing and decide the process sequence (process planning).
  - Decide the machines, cutting tools, work-holding devices and gauges required.
  - Estimate the cost of tooling and the pay-back (economic justification).
  - Design the tools, prepare drawings and supervise manufacture and try-out.
  - Maintain, repair, recondition and reclaim the tooling.

1.3 Tool Classification

- Tools can be grouped by function as shown in the chart below.



Group	Purpose	Examples
Cutting tools	remove metal as chips	single point tool, milling cutter, drill, reamer, broach, tap
Work-holding	locate and clamp the work	drill jigs, milling/boring fixtures, chucks, vices
Press tools	cut or form sheet metal	blanking, piercing, bending, drawing dies
Gauges	check size, shape, position	plug gauge, ring gauge, snap gauge, indicating gauge
Special tools	assist assembly/forming	moulds, forging dies, welding fixtures, assembly tools

- Another classification: standard tools (available off-the-shelf, e.g. drill bits) and special tools (designed for one particular job, e.g. a jig for a casing).

- Based on production volume: general-purpose, semi-special, special-purpose (dedicated) tooling.

2. Tool Design Objectives

- To reduce the manufacturing cost per piece (lower labour and machining time).
- To increase the production rate by reducing set-up, loading and idle times.
- To maintain the quality and accuracy : parts must be within tolerance and interchangeable.
- To make the operation safe and easy - reduce operator fatigue, guard moving parts.
- To utilise machine tools fully and allow semi-skilled labour to do skilled work.
- To design tools that are simple, rigid, light, durable and easy to maintain / repair .
- To use standard parts wherever possible (bushes, screws, pins, clamps) to cut cost and lead time.
- To allow the use of less costly material or process by clever tooling.

Tool design = Economy + Accuracy + Safety + Productivity

- Design must balance tool cost against number of parts to be produced
- Break-even quantity: N = (tool cost) / (saving per piece)

3. Tool Design in Manufacturing

3.1 General tool design procedure

1. Statement of the problem - study part drawing, material, quantity, tolerances, machine available.
2. Analysis of the problem - operations needed, locating surfaces, clamping, forces involved.
3. Analysis of the requirements - accuracy, loading/unloading, chip disposal, cost limit.
4. Development of possible solutions - make several rough sketches (ideas).
5. Evaluation of the ideas - compare on cost, accuracy, time, safety; pick the best.
6. Design and detail - calculate forces, select materials, prepare assembly and detail drawings.
7. Prototype / try-out and testing , then modification.
8. Final drawings, bill of materials, release to the tool room and documentation.

3.2 Place of tool design in the manufacturing cycle

- Product design -> Process planning -> Tool design -> Tool manufacture -> Production -> Inspection.
- Tooling is a link between the product designer's drawing and the finished part.
- Good tooling affects: lead time, cost, quality, scrap, safety and flexibility.

3.3 Factors affecting tool design

- Part: size, shape, material, tolerance, surface finish, lot size.
- Machine: type, size, power, speed, table size, accuracy, available.
- Production: rate required, manual / automatic, shop skill, tool-room capacity.
- Economic: cost of tool, life of tool, delivery time, pay-back period.

4. Challenges and Requirements

4.1 Challenges in modern tool design

- Shorter product life cycles -> tooling must be ready faster (rapid tooling, CAD/CAM).
- Tight tolerances and better finish demanded by customers.
- New hard-to-machine materials (titanium, Inconel, composites) need new tool geometry and materials.
- High speed machining causes heat, vibration and tool wear problems.
- Flexibility : small batches and many variants -> modular and quick-change tooling.
- Cost pressure and need for lean manufacturing, less scrap.
- Skilled tool makers are scarce; need automation, CNC, EDM, 3D printing of tooling.
- Environmental rules: cutting fluids, dry machining, recycling of tools.

4.2 Basic requirements of a good tool

- Sufficient strength and rigidity to resist forces without deflection.
- Accuracy and repeatability of location of every workpiece.
- Wear resistance and hardness at the working edges; toughness against shock.
- Easy and quick loading and unloading, with fool-proof design (the part can be placed only one way).
- Provision for chip clearance and removal; coolant access.
- Low cost , interchangeable standard parts, easy maintenance and repair.
- Adequate safety - no sharp corners, guards, clamp that cannot slip.
- Good appearance and finish , proper marking and ease of handling (weight, handles, lifting hooks).

5. Standards in Tool Design

- Standardisation = using agreed sizes, shapes, materials and practices so that parts can be bought, replaced and interchanged.
- Advantages: lower cost, shorter lead time, easy replacement, better quality, fewer varieties in stores.
- Standard bodies: BIS (IS) - India, ISO - international, ANSI/ASME - USA, DIN - Germany, BS - Britain, JIS - Japan.

Standard	Used for
IS 696 / SP 46	engineering drawing practice (lines, views, dimensioning)
IS 919 / ISO 286	limits and fits for shafts and holes
IS 3073	assessment of surface texture (Ra values)
IS 2362 / ISO 2768	general tolerances (no tolerance stated)
IS 3403 / ISO	standard drill bushes, press tool parts, die sets

- Standard components in tooling: drill bushes, locating pins, clamps, T-bolts, nuts, springs, punches, die sets, dowel pins, screws .
- Always choose a standard part first; design a special part only if no standard one works.

6. Tool Drawings

- A tool drawing is the "language" by which the designer tells the tool maker how to make and assemble the tool.
- Types of tool drawings:
  - Layout / concept sketch: freehand first idea.
  - Assembly drawing: shows all parts in position with a bill of materials (BOM) and balloons.
  - Detail (part) drawing: each part separately with full dimensions, tolerances, material, heat treatment, finish.
  - Sub-assembly drawing for complex tools.

Items on a tool drawing

- Title block: tool name, part no., scale, material, drawn / checked by, date
- Views: front, top, side, sections - as few as are enough to show the shape
- Dimensions with tolerances , surface finish symbols, fits
- Notes: heat treatment (e.g. harden 58-62 HRC), grind all over, deburr, marking
- Bill of materials: item no., description, qty, material, standard / purchased
- Part drawing of the workpiece shown in chain lines for reference (in jigs/fixtures)

- Rules to follow:

1. Use BIS / ISO conventions - first angle projection (India), standard line types, scale 1:1 where possible.
2. Show only the necessary tolerances - tight tolerance costs more.
3. Give dimensions from datum (locating) surfaces to avoid tolerance build-up.
4. Mention material and hardness for every part.
5. Leave clear space for clamping, chip removal and operator hands .

7. Surface Finish

- Surface finish (roughness) is the fine irregularity left on the surface by the manufacturing process.
- Measured by Ra (CLA) = arithmetic average of the heights of peaks and valleys from the mean line, in micrometre (um).
- Ra = (1/L) x integral of |y| dx, over the sampling length L.
- Finish affects wear, friction, fatigue strength, corrosion, appearance, sealing and fit quality.

Process	Ra (um)	Typical tooling use
Rough turning / milling	3.2 - 12.5	non-working surfaces of bodies
Finish turning / milling	0.8 - 3.2	base of jigs, fixture bodies
Surface grinding	0.4 - 1.6	locating faces, die blocks
Cylindrical grinding	0.2 - 0.8	pins, punches, bush bore
Honing / lapping	0.05 - 0.4	gauges, bush holes, dies
Polishing	0.025 - 0.2	forming and drawing die surfaces

- Rule: the finer the finish, the higher the cost - so specify only what the function needs.
- Roughness symbol: a check mark with the Ra value written on it; N-grade (N1 to N12)

also used (N6 = Ra 0.8 um).

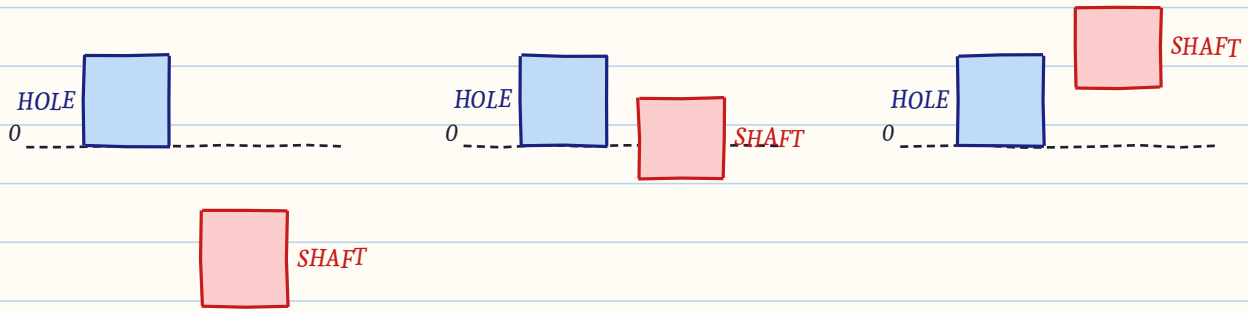
- Typical specifications in tools: locating surfaces Ra 0.8 , punch/die working surfaces Ra 0.4 or better , drill-bush bore Ra 0.4 .

8. Fits and Tolerances

8.1 Basic terms (IS 919 / ISO 286)

- Basic size: nominal size from which limits are measured.
- Limits of size: maximum and minimum permissible sizes.
- Tolerance = upper limit - lower limit (total permitted variation).
- Deviation: difference between actual/limit size and basic size. Fundamental deviation fixes the position of the tolerance zone.
- Allowance: intentional difference between mating parts (minimum clearance or maximum interference).
- Clearance = hole size - shaft size (positive). Interference = shaft - hole (positive).
- Unilateral tolerance: both deviations on one side of basic size. Bilateral: on both sides (e.g. 25 +/- 0.02).

8.2 Types of fits



Clearance: shaft always smaller | Transition: may be either | Interference: shaft always larger

Basic hole system: hole lower deviation = 0 (H), e.g. H7/g6 (clearance), H7/k6 (transition), H7/p6 (interference)

Fit	Condition	Tooling example
Clearance (running/sliding)	hole always larger than shaft	punch in guide, bush & liner sliding, H7/g6
Transition (location)	clearance or interference	dowel pin position, locating pin, H7/k6, H7/n6

Fit	Condition	Tooling example
Interference (press)	shaft always larger than hole	bush pressed into jig plate, liner, H7/p6, H7/r6

8.3 Systems of fits

- Hole basis system: hole size fixed (H, lower deviation = 0); shaft size varied to get the fit. Preferred - holes are made by standard drills/reamers, so fewer tools.
- Shaft basis system: shaft fixed (h, upper deviation = 0); hole varied. Used where one shaft carries many parts (e.g. line shafts).
- Tolerance grades IT01 to IT16: smaller number = finer tolerance. Gauges IT1-IT5, fits IT5-IT12, rough work IT12-IT16.

Typical fits in tool design

- Bush press fit in jig plate: H7/p6 (or H7/n6)
- Renewable (slip) bush in liner: H7/g6 or F7/h6
- Drill running fit in bush bore: F7
- Dowel pin in tool body: H7/m6, H7/n6
- Punch in stripper guide: H7/h6 (sliding)
- Punch and die: not a standard fit - clearance fixed by material and thickness

- Tolerance stack-up: total tolerance on a chain of dimensions = sum of individual tolerances -> keep chains short.
- Rule of thumb: tool tolerance should be about 10 - 20 % of the part tolerance (a tool 5 to 10 times more accurate than the part).

9. Tooling Materials

- Tool materials should have: hardness, wear resistance, toughness, hot hardness (red hardness), shock resistance, machinability and grindability, low distortion in heat treatment, availability and low cost.

9.1 Ferrous tooling materials

Material	Composition / Properties	Used for
Grey cast iron	C 3-3.5 %; good damping, casting, cheap	jig and fixture bodies, die shoes
Mild steel (IS 1570: 10C4, 20C8)	low C; easy machine, weld; low hardness	plates, bases, clamps; carburised parts

Material	Composition / Properties	Used for
Medium carbon steel (C40, C45)	0.4-0.5 % C; tough; hardenable moderate	bodies, shanks, shafts, pins
Plain carbon tool steel (T55 - T130)	0.6-1.3 % C; hard after quench; poor hot hardness (<200 C)	taps, files, punches, hand tools, simple dies
Case-hardening steels (15Ni..., En-36)	tough core, hard case	locators, bushes, pins, jig plates
Alloy tool steel (cold work: D2, O1)	Cr, Mn, W, Mo give deep hardening, low distortion	blanking dies, punches, gauges
Hot work tool steel (H11, H13)	Cr-Mo-V; hot hardness to 550 C	forging dies, extrusion, die casting
High speed steel (HSS)	18-4-1 (T1) or M2, M42 (Co); red hardness 600 C	drills, milling cutters, reamers, broaches
Stainless steel	Cr 12-18 %; corrosion resistant	food / medical tooling

- *HSS composition:* T1 = 18 % W, 4 % Cr, 1 % V; M2 = 6W-5Mo-4Cr-2V; adding 5-10 % Co (M42) gives higher hot hardness.
- *Ranking by hot hardness:* carbon tool steel < HSS < cast alloy < carbide < ceramic < CBN < diamond.

9.2 Non-ferrous tooling materials

- *Aluminium alloys:* light weight, easy to machine (use for large fixtures, hand-held jigs, prototypes); low wear resistance.
- *Brass and bronze:* good bearing material, non-magnetic, non-sparking; used for bushes, wear strips, liners.
- *Phosphor bronze, gunmetal:* bearings, bushes and worm-wheels in tooling.
- *Copper alloys (Be-Cu):* electrodes for EDM, spot welding tips, non-sparking tools.
- *Zinc alloys (Kirksite):* low-cost short-run drawing dies and moulds; easy casting.
- *Cast non-ferrous alloys / Stellite* (Co-Cr-W): tough, hot hard 800 C, cutting tips, wear faces.
- *Lead and low melting alloys:* used as soft jaws or to hold thin parts (pouring around the part).
- *Titanium / Tungsten / Molybdenum:* special high temperature tools.

9.3 Carbides, Ceramics and Diamond

- Cemented carbides: WC (tungsten carbide) grains bonded with Co by powder metallurgy (sintering).
  - Very hard (HRA 89-93), keeps hardness to ~ 1000 C; speeds 3-5 times HSS.
  - Grades: P (steel cutting; WC + TiC + TaC), M (multi-purpose), K (cast iron, non-ferrous; plain WC-Co).
  - More cobalt = tougher but softer. Brittle - needs rigid set-up. Used as brazed tips or indexable inserts , also coated (TiN, TiC, Al2O3).
  - In dies: carbide inserts for long-run blanking, drawing dies, bush liners, gauge wear faces.
- Ceramics: Al2O3 (alumina) with some ZrO2, or Si3N4, SiAlON.
  - Hardness to 1200 C, chemically inert; very high speeds on cast iron, hardened steel; but low toughness - need rigid, no interrupted cuts.
- Cubic Boron Nitride (CBN): hardness next to diamond; used for hardened steels (>45 HRC), stable to 1300 C and does not react with iron.
- Diamond (natural, synthetic PCD):
  - Hardest known material; cuts aluminium, copper, plastics, composites, graphite, glass with excellent finish.
  - Not used on ferrous metals - carbon diffuses into iron at high temperature; oxidises above ~ 700 C.
  - Also used for dressing grinding wheels, wire-drawing dies, lapping .

9.4 Non-metallic tooling materials

- Plastics: nylon, Teflon (PTFE), acetal (Delrin), bakelite, epoxy, fibre-glass - used for soft jaws, locator pads, wear pads, liners, light jigs ; they do not mar the finished surface.
- Epoxy / resin tooling: cast epoxy with filler (aluminium powder) for prototype dies, drilling jigs, check fixtures ; quick and cheap.
- Rubber and polyurethane: strippers, pressure pads, forming dies (Guerin process, rubber-pad forming), springs and cushions.
- Wood and laminates: patterns, templates, cheap fixtures for small quantity.
- Granite and ceramics: surface plates, bases of measuring tools (stable, no rust, no distortion).
- Cermets, graphite: EDM electrodes, hot forming die inserts.
- Advantages: light, cheap, corrosion-free, easy to make, good damping; disadvantage: low

strength and wear resistance, dimensional instability with moisture or temperature.

10. Designing with Relation to Heat Treatment

- Heat treatment changes the properties (hardness, toughness, grain) of steel by heating and cooling. Tool components are normally hardened and tempered .

Process	What is done	Purpose
Annealing	heat above critical temp, furnace cool	soften, relieve stress, improve machinability
Normalising	heat above critical, air cool	refine grain, uniform structure
Hardening	heat to austenite, quench in oil/water/air	maximum hardness
Tempering	reheat to 150-650 C after quenching	reduce brittleness and stress, adjust hardness
Case hardening	carburise / nitride / cyanide the surface	hard wear-resistant skin, tough core
Stress relieving	heat 500-650 C, slow cool	remove machining stress

10.1 Design rules to avoid heat treatment failure

- Distortion and cracking arise from uneven heating/cooling and sudden changes of section. So:

1. Keep the section thickness uniform ; avoid thick-thin joins.
2. Provide generous fillets and radii (min. 3 mm) at corners - sharp internal corners are crack starters (stress concentration).
3. Avoid sharp edges and notches ; chamfer or round the edges.
4. Avoid blind holes and holes near edges - quench liquid gets trapped (use through holes; plug with fire clay if blind).
5. Symmetrical shapes distort less; balance holes and slots.
6. Keep holes away from edges - at least hole diameter (min. 1.5 x dia.) of material around it.
7. Leave grinding allowance (0.2 to 0.5 mm) on hardened surfaces for finishing after heat treatment.
8. Where possible, use sectional (built-up) construction - a large complex die is made from small hardened blocks, easy to harden, grind and replace.

9. Rough machine -> stress relieve -> finish machine -> harden -> temper -> grind.
10. Choose the right steel : oil-hardening (O1) or air-hardening (A2, D2) for complex shapes (less distortion), water-hardening only for simple shapes.

Remember (exam points)

- Hardness of punches/dies: 58-62 HRC ; die inserts of D2: 60-62 HRC
- Jig bushes / locators: 58-62 HRC (case hardened, depth 0.8 - 1.5 mm)
- Springs and clamps: tempered to 40-50 HRC
- Jig bodies (cast iron / MS): normally not hardened but stress relieved
- Quench cracks: sharp corners, sudden thickness change, wrong quenchant

- Decarburisation (loss of carbon at the surface) is prevented by controlled atmosphere, vacuum or salt bath heating, or by leaving grinding allowance.

11. Quick Revision - Unit I

- Tool design -> link between product drawing and production; objectives = quality, cost, productivity, safety .
- Tool classes: cutting, work-holding (jigs-fixtures), press tools, gauges .
- Standards: IS, ISO, DIN, BS; use standard parts.
- Fits: H7/p6 press, H7/k6 transition, H7/g6 clearance ; hole-basis system preferred.
- Materials: C.I. (bodies), MS (plates), tool steels (dies), HSS (cutters), carbide / ceramic / CBN / diamond (hard machining).
- Heat treatment design: uniform sections, radii, no sharp corners, grinding allowance.